SCENE-YOLO: A One-Stage Remote Sensing Object Detection Network With Scene Supervision

Tao Zhao, *Student Member, IEEE*, Ruyi Feng, *Member, IEEE*, and Lizhe Wang, *Fellow, IEEE*

***Abstract*— The ground object distribution in remote sensing images exhibits strong regularities. However, existing deep learning-based object detection models often focus solely on instance-level information within sample labels, neglecting the modeling of relationships between instances. Additionally, these models fail to effectively utilize the substantial background information present in remote sensing images. To address these issues, we propose a remote sensing object detection network named SCENE-YOLO, building upon the YOLOv8 architecture and introducing scene supervision. First, we introduce a scene information gathering and distribute network (SGD), based on transformer, to inject high-level semantic information into the feature pyramid. A slice-and-distribute mechanism is employed to prevent information loss during feature fusion across layers. Second, the backbone network is redesigned, incorporating the attention mechanism of omni-dimensional dynamic convolution (ODConv) to dynamically redistribute weights for target features. Subsequently, a scene label generation algorithm (SLGA) based on prototype learning is proposed to supervise the model by generating scene-level labels, modeling instance–instance relationships through the introduction of artificial knowledge and multilevel classification. Finally, a scene-assisted detection head (SADHead) is introduced to enhance detection performance in complex backgrounds by leveraging scene features with global contextual information to assist the model in target classification. Experimental validation on the publicly available DOTA and DIOR datasets demonstrates the effectiveness and superiority of the proposed algorithm.**

***Index Terms*— Deep learning, object detection, remote sensing images.**

I. Introduction

REMOTE sensing technology is a scientific and technological discipline that utilizes satellites, aircraft, or other distant sensors to acquire information about the Earth's surface. Initially, remote sensing relied predominantly on photogrammetry, analyzing aerial or satellite images to obtain surface information. However, with continuous innovations in various sensor technologies, remote sensing now encompasses a wide range of electromagnetic radiation bands, including visible light, infrared, and microwaves, resulting in more comprehensive and diverse Earth information acquisition. As a substantial volume of remote sensing data accumulates, the accurate and efficient identification and localization of specific targets from massive remote sensing images have become pressing challenges.

The task of remote sensing object detection aims to leverage advanced technologies such as computer vision and deep learning to automatically identify and extract points of interest from remote sensing images, including structures, vehicles, natural disasters, and more. This task holds significant importance in areas such as urban planning [1], resource management [2], and environmental monitoring [3], [4]. Consequently, the development of efficient and accurate remote sensing object detection algorithms has become a crucial direction in the advancement of remote sensing technology. Compared to other types of remote sensing images, optical remote sensing images exhibit superior characteristics in target detection tasks, such as high resolution, ease of observation and interpretation, advantages in detecting radar-stealth military targets, and strong feature extraction capabilities. Consequently, they are better suited for remote sensing target detection tasks. In this article, all references to remote sensing images specifically refer to optical remote sensing images.

Object detection technology can be broadly categorized into two main types: traditional manual methods [5] and deep learning-based approaches [6], [7], [8]. Traditional manual methods emphasize the design of target feature extraction techniques. However, due to the challenging task of diverse requirements, the applicability of these methods is typically confined to specific scenarios [9]. In contrast, with the increasing availability of extensive datasets and advancements in computational technology, deep learning-based methods have gained significant popularity in the field of object detection. Compared to traditional manual object detection techniques, deep learning-based methods offer faster detection speeds and higher accuracy. Consequently, deep learning-driven object detection is rapidly evolving, demonstrating the capability to efficiently and accurately detect targets and extract scene information from remote sensing images.

Simultaneously, remote sensing object detection faces a series of formidable challenges, including the complexity of image backgrounds, the diversity of ground objects, the presence of small and densely distributed targets, and the inconsistency in scale [10]. These challenges pose significant difficulties for remote sensing object detection. However, in comparison to natural images, remote sensing images possess unique characteristics. Due to the high altitude at which remote sensing images are captured, ground objects appear small, with the background region occupying the vast majority

Received 30 September 2024; revised 8 January 2025; accepted 30 January 2025. Date of publication 7 February 2025; date of current version 18 February 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 41925007 and Grant 42471430, in part by the Knowledge Innovation Program of Wuhan-Shuguang under Grant 2023010201020336, in part by Hubei Natural Science Foundation under Grant 2024AFB561, and in part by the Funds for International Cooperation and Exchange of the National Natural Science Foundation of China under Grant 42311530065. *(Corresponding author: Ruyi Feng.)*

The authors are with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: zhaotao123@cug.edu.cn; fengry@cug.edu.cn; lizhe.wang@gmail.com).

Digital Object Identifier 10.1109/TGRS.2025.3539698

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

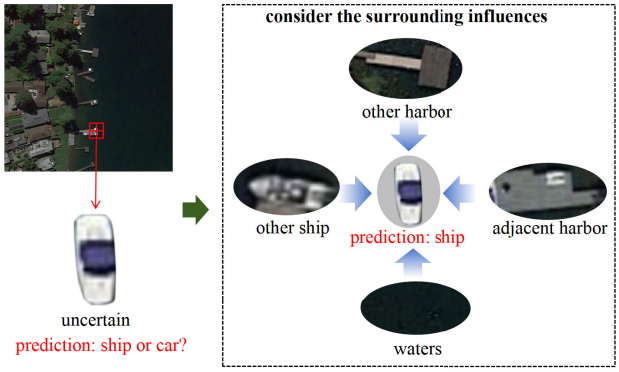


Fig. 1. Influence of surrounding objects on target detection.

of the image. Additionally, the distribution of ground objects in remote sensing images exhibits stronger regularities; similar categories of ground objects often cluster together, showing a strong correlation with the background. For instance, ships and harbors frequently co-occur in waterfront areas, and various types of aircraft are commonly found in airport regions. As shown in Fig. 1, the surrounding objects in the vicinity of a target can provide highly valuable contextual information for target classification, thereby assisting the model in generating more accurate predictions.

Considering these aspects, numerous scholars have proposed remote sensing target detection methods that incorporate contextual information. For instance, Zhang et al. [11] proposed a context-aware detection network (CAD-Net), which integrates contextual information and attention mechanisms into classical anchor-based models. This approach enables implicit learning of the relationships between anchors and their surroundings but results in uncontrollable outcomes and limited interpretability. Similarly, Tao et al. [12] introduced a scene classification module to incorporate scene priors into vehicle detection. While this method achieved some progress, it suffers from poor generalizability due to its task-specific design. Qian et al. [13] incorporated a scene branch into a two-stage detection model, leveraging global pooling layers and fully connected layers to learn global scene features from image characteristics. This enhancement improved detection performance by refining classification results. However, despite extracting and utilizing scene information, this approach does not adequately differentiate scene features and fails to address the challenges posed by implicit learning.

To enhance the interpretability of scene prediction and the controllability of training, together with addressing the optimization challenges of implicit learning models, explicit scene supervision signals can be employed to guide the scene classification task. Zhang et al. [14] utilized scene-level labels and trained convolutional weights with strong discriminative ability through mutual cues between scene pairs, achieving promising results in weakly supervised object detection. This demonstrated the feasibility of leveraging scene information to improve instance-level object classification performance.

However, due to the extensive field of view in remote sensing images, a single image may simultaneously contain multiple different scenes, making the scene classification task inherently a multilabel classification problem. Some scholars have addressed multilabel classification tasks using prototype learning approaches. Wang et al. [15] proposed a subspace prototype learning method for few-shot remote sensing scene classification. During the meta-test phase, a subspace prototype was learned for each class of samples, and classification was achieved by measuring the distance between samples and prototypes. Zhou et al. [16] introduced a framework called category prototype compositional learning (CPCL), which modeled the compositional characteristics of multilabel images using prototype learning combined with the semantic representation of category prototypes. They employed transformed appearance distance (TAD) for metric learning, thereby enhancing the performance of multilabel image classification.

These prototype-based learning methods handle category correlations and predict multilabels by defining class centers or prototypes and using metric-based approaches. However, such methods rely heavily on the accurate representation of class features and are susceptible to issues such as sensitivity to noise and outliers, as well as overlapping distributions between classes.

In this article, we address the unique characteristics of remote sensing images and propose the SCENE-YOLO, a one-stage remote sensing object detection network with scene supervision. Specifically, SCENE-YOLO introduces the scene information gathering and distribution network (SGD), which receives feature maps from various levels of the backbone network. Leveraging the superior ability of transformers to capture global contextual information, the SGD network extracts scene-level features while effectively distinguishing them from instance-level features. Simultaneously, it models the long-distance dependencies between instances and their surrounding environments. The features extracted by SGD are fused losslessly with the feature maps output by feature pyramid network (FPN) through a collection distribution mechanism [17]. Additionally, we abstract the scene of an image as a probability distribution over a set of scene prototypes. The scene-assisted detection head (SADHead) encodes the fused features into a scene prediction vector, representing the probability distribution of scenes for each image sample. Furthermore, to provide supervisory signals for scene prediction, a prototype-based scene label generation algorithm (SLGA) is proposed. By incorporating human prior knowledge through parent–child category divisions and sample selection, the algorithm generates more discriminative class features. In summary, our contributions are as follows.

1) We propose the scene-assisted detection framework, which incorporates a scene prediction branch into the traditional detection framework, making it more suitable for object detection in remote sensing images.
2) We reevaluate the limitations of the FPN and introduce a feature fusion network based on the gather and distribute mechanism. This enhances the SCENE-YOLO model's capability to learn scene features effectively.
3) We improve the conventional C2f module by introducing omni-dimensional dynamic convolution (ODConv) [18], providing the backbone with complementary multidimensional attention to focus on more crucial target areas.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

4) A robust SLGA and a novel detection head. Leveraging the scene supervision signal generated by SLGA, the model learns to predict the scene of the image, utilizing the prediction results to assist in object category detection.

II. Related Works

A. Object Detection

The object detection task encompasses two primary aspects: classification and regression. Currently, target detection algorithms can be broadly categorized into two major classes: two-stage algorithms and one-stage algorithms.

1) Two-Stage Algorithm: It divides the target detection task into two critical stages: candidate region extraction and target classification with localization. Initially, the algorithm generates a series of candidate regions that may contain the target in the candidate region extraction stage. Subsequently, these candidate regions are fed into the second stage for target classification and localization. In this stage, deep neural networks are commonly employed for feature extraction from candidate regions, performing target classification, and precisely locating bounding boxes. The region with CNN (R-CNN) [19] represents the paradigm of two-stage target detection, extracting candidate proposals from input images and subsequently processing them using deep CNN. To improve detection efficiency, Fast R-CNN [20] was introduced, integrating the entire detection process into a convolutional neural network by introducing region of interest pooling. Faster R-CNN [21] further refined the candidate region extraction stage by introducing the region proposal network (RPN), trained end-to-end, thereby enhancing accuracy. Mask R-CNN [22] extends upon Faster R-CNN, not only facilitating target detection and localization but also enabling instance segmentation, generating precise pixel-level masks for identified objects. This extension provides robust support for more complex scene understanding. To enhance the detector's adaptability to various targets, Cascade R-CNN [23] introduces multiple cascaded subnetworks during the training process. Each subnetwork focuses on addressing samples of varying difficulty levels.

2) One-Stage Algorithm: The development history of single-stage object detection algorithms is relatively short, but significant progress has been made in recent years [24], [25]. These algorithms primarily accomplish object detection tasks through a single forward pass, offering faster processing speeds compared to two-stage algorithms.

The you only look once (YOLO) [26]series is a classic representation of single-stage object detection algorithms. By dividing an image into grids and simultaneously predicting the positions and categories of multiple bounding boxes within each grid, it reformulates object detection as a regression problem. YOLOv8, developed by Ultralytics in 2023 based on YOLOv5, is the latest version in the YOLO series. Since its release, the model has been applied to over 12 million object detection tasks, demonstrating its practicality.

Fig. 2 shows the architecture of YOLOv8, which comprises three key modules: the backbone network, the feature extraction network, and the detection network. YOLOv8 adopts

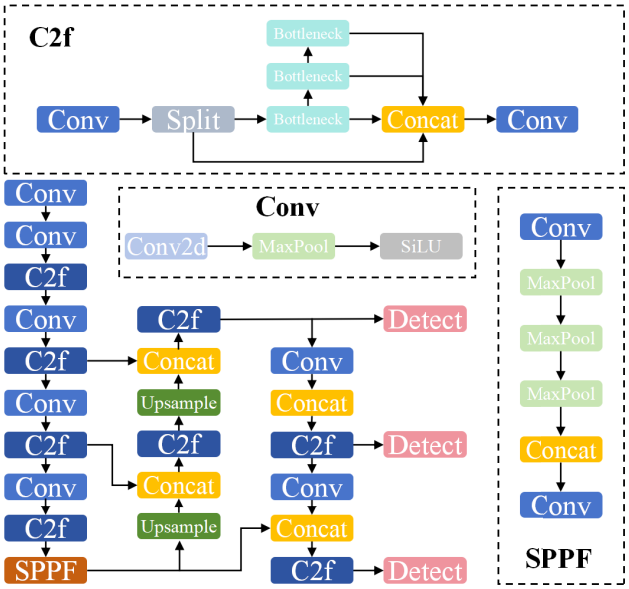


Fig. 2. YOLOv8 network architecture.

CSPDarknet53 as its backbone, which improves upon VGG-16 by incorporating a cross-stage partial (CSP) architecture [27], significantly enhancing detection accuracy and efficiency. CSPDarknet53 achieves computational efficiency by splitting residual blocks into a main branch and a side branch. The main branch performs convolution operations and feature extraction, while the side branch reduces the number of channels through 1×1 convolution [28].

The open-source version of YOLOv8, available on GitHub, includes five variants: YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x. These variants control model complexity by adjusting depth and width multipliers. Detailed model parameters are presented in Table I.

Apart from the YOLO series, several other single-stage algorithms are also representative in this domain. For instance, the single shot multibox detector (SSD) [29] detects objects of different sizes and ratios by predicting bounding boxes on multiple feature maps at various levels. RetinaNet [30] extends SSD by addressing class imbalance through the use of focal loss. EfficientDet [31] employs the compound scaling strategy, which, while reducing model complexity, maintains high detection accuracy.

B. Context Modeling in Object Detection

Context refers to the spatial and semantic relationships between the target of interest and its surrounding environment. Numerous empirical studies on general object detection in remote sensing images [32], [33] suggest that object detectors can enhance detection performance for challenging weak targets by mining relevant contextual information and appropriately modeling instance–instance and instance–scene relationships.

Currently, methods can be categorized based on different types of contextual information into local contextual information mining and global contextual information mining. Zhang et al. [34] scaled the original region proposal at different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
GENERAL COMPARISON OF YOLOV8 MODELS

Model	Depth	Width	Max channels	Params	FLOPS
YOLOv8-n	0.33	0.50	1024	3.2M	8.7B
YOLOv8-s	0.33	0.50	1024	11.2M	28.6B
YOLOv8-m	0.67	0.75	768	25.9M	78.9B
YOLOv8-l	1.00	1.00	512	43.7M	165.2B
YOLOv8-x	1.00	1.25	512	68.2M	257.8B

scales to obtain three local context regions, and designed a context dual enhancement module to fuse context features and object-level features. Zhao et al. [35] designed a scene contextualized detection Network (SCDNet), which can enhance the exploration of the relationship between small objects and their surroundings by decoupling scene context information through a dedicated scene classification subnet (SCN). Gong et al. [36] proposed a context-aware convolutional neural network (CA-CNN), incorporating a context-regions-of-interests (Context-RoIs) mining layer to learn contextual information around objects. And by merging a series of region proposals around the target to generate contextual RoIs for fusion with object-level RoIs.

Considering the superior performance of graph convolutional networks (GCNs) [37] in modeling object relationships, Tian et al. [38] enhanced feature discriminability by introducing semantic information. They employed two branches of GCNs to learn spatial semantic graph similarity, achieving excellent results in multiclass geographical spatial target detection. The aforementioned methods are based on local contextual information mining, while global contextual information establishes connections between objects and scenes on a broader scale. For instance, in ports, ships and airplanes are typically found parked at airports. Chen et al. [39] used RoI alignment operations on global feature maps to capture scene-level contextual information. They then fused these features with object-level RoI features to strengthen the correlation between objects and scenes. Zheng et al. [40] proposed a foreground relation-aware network (FarSeg), which employs a scene encoding branch to associate foreground with relevant context, enhancing the disparity between foreground and background features. Liu et al. [41] designed a scene FPN network, which sets up two branches to learn scene-level features and object-level features, respectively, and fuses them at different scales. Then, they use one hot encoding-based scene-level supervision to generate predicted scene vectors to assist in model classification. The aforementioned approaches alleviate the complexities and noise interference in remote sensing images in diverse manners, enhancing the model's sensitivity to target regions and improving the performance of object detection in remote sensing images. However, challenges persist in the underutilization of both local and global contextual information. Additionally, in explicitly supervised methods, the generation of scene-level auxiliary labels fails to guarantee high quality. In certain scenarios, it may even misguide the model training, rendering it ineffective in addressing the challenges posed by insufficient feature extraction for small and weak targets.

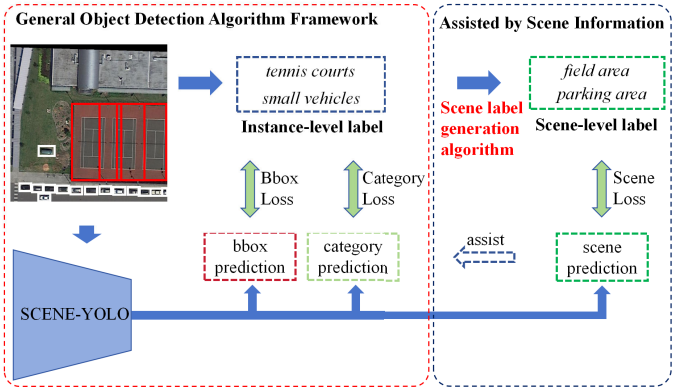


Fig. 3. Scene-assisted detection framework.

III. PROPOSED METHOD

In this section, we present our SCENE-YOLO for precise remote sensing object detection, which is based on YOLOv8. We initially introduce the scene-assisted detection framework and the overall network structure of SCENE-YOLO in Section III-A. Building upon this framework, SGD and ODC2f are proposed in Sections III-B and III-C, respectively, to enhance the detector's capability to capture global features. Subsequently, we elaborate on our proposed SLGA in Section III-D and demonstrate how the model utilizes scene labels to assist classification and improve detection performance through the introduction of SADHead in Section III-E.

A. Scene-Assisted Detection Framework

As mentioned above, the distribution of target features in remote sensing images exhibits two important characteristics: aggregation of similar categories and a strong correlation between targets and the background. Therefore, to integrate the characteristics of target detection frameworks and remote sensing images, recent methods have employed carefully designed structures to extract scene information [42], [43]. In [43], an enhanced feature extraction method named SMENet is proposed to maintain the consistency of spatial and semantic information during the feature fusion process.

Differing from the aforementioned methods that all involve redesigning a feature extraction network to capture global scene features, it comes with a high computational cost and does not effectively leverage the existing detection framework. Considering that in computer vision tasks, the same backbone can be fine-tuned for different downstream tasks, as shown in Fig. 3, to maximize the utilization of the existing detection framework and simplify the network structure, we propose the scene-assisted detection framework.

This framework introduces a scene-assisted component onto traditional object detection algorithm frameworks. In addition to predicting bounding boxes and classifications for input images, the detector also predicts the scene of the image. The scene prediction provides a classification bias term to assist classification predictions and calculates loss in conjunction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

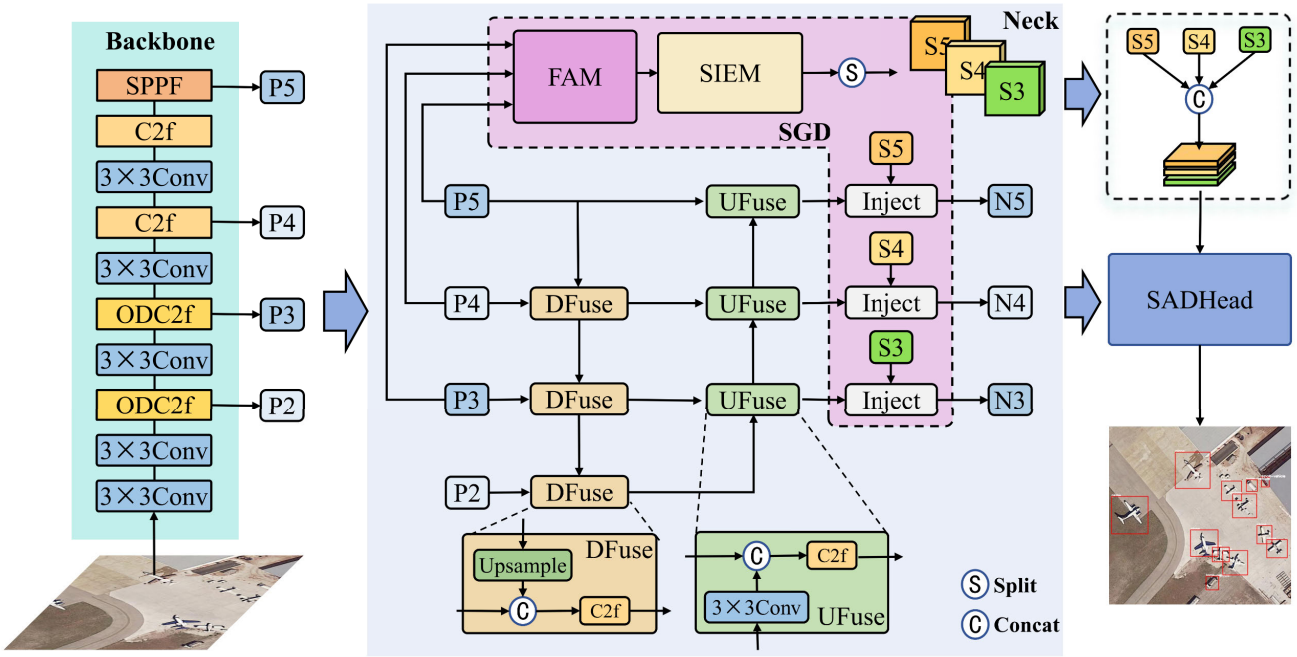


Fig. 4. Network architecture of SCENE-YOLO.

with scene-level labels obtained from SLGA. Sections III-D and III-E describe these components in detail.

Fig. 4 illustrates the overall structure of the proposed SCENE-YOLO. To minimize redundant computations and achieve efficient utilization of the feature extraction network, scene classification and object detection share the same backbone. Scene feature extraction is carried out only in the neck part of the network through SGD. This implementation allows for the extraction of scene features with only a marginal increase in computational cost, accomplishing the integration of the object detection framework and remote sensing scene classification and we also add a more fine-grained detection layer P_2 in the FPN section to enhance the spatial detail features of small targets. Additionally, we have made certain improvements to the backbone, enabling it to dynamically respond to inputs, thereby enhancing the flexibility and robustness of the backbone for both tasks. These enhancements will be detailed in Sections III-B and III-C.

B. Scene Information Gathering and Distribution Network

Previous methods typically employed FPN to fuse multi-scale local contextual features of targets, detecting objects based on local spatial features. Although in FPN, high-dimensional semantic information is propagated downward, the semantic information tends to be progressively lost in the layer-wise fusion process. This results in a suboptimal utilization of these scene-level features in FPN, particularly in low-dimensional, high-resolution feature layers. Consequently, the model may exhibit lower accuracy in detecting objects in certain scenes, especially in remote sensing images with numerous small and weak targets [17]. In reality, the category of detected objects is often closely related to the scene they are in. For example, an inconspicuous rectangular object is more likely to be a ship if it is on the sea and more likely to be a small car if it is on a road. Du et al. [44] employed point-wise convolution to generate global contextual features and designed a context-enhanced group normalization (CEGN) layer to integrate global contextual information into sparse features, avoiding inaccuracies caused by interpolation algorithms [45]. However, the ability of point-wise convolution to extract background information still depends on the receptive field of the original network and cannot capture long-distance dependencies. Nevertheless, point-wise convolution has the advantage of low computational cost, meeting the performance requirements of real-time algorithms and demonstrating excellent performance in real-time UAV image object detection tasks. When birds fly far away, they appear as small black dots, but we can infer they are birds based on the background of the sky. Motivated by this observation, Xiao et al. [46] attempted to leverage the background surrounding small objects to assist in perceiving the objects themselves. They proposed a context enhancement module (CEM), which processes the original features through multiple branches with different convolution rates, extracting features with varying receptive fields. This approach achieved performance improvements in small object detection. However, fixed receptive fields still suffer from poor generalization, particularly when dealing with large objects and densely packed targets.

Inspired by this, we have devised a scene information gather module (SIGM) (shown in Fig. 5), which employs a feature alignment module (FAM) to standardize the spatial dimensions of feature maps across different scales. Subsequently, it takes the unified features as input to the scene information extraction module (SIEM), a transformer-based module, to extract globally contextualized high-dimensional scene information from feature maps at various levels of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

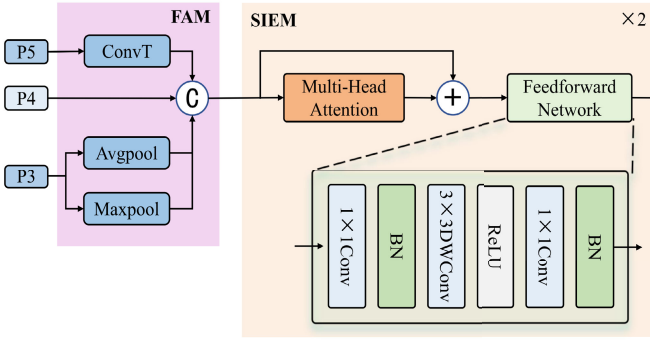


Fig. 5. SIGM structure diagram.

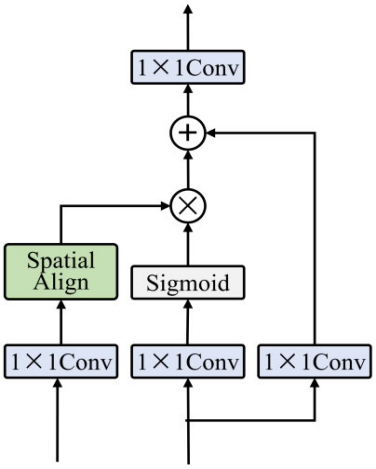


Fig. 6. Inject structure diagram.

backbone. This process enables the effective utilization of background regions, effectively leverages surrounding objects and background information. This is because, when computing multihead attention, the feature map undergoes correlation calculations, establishing relationships between features at different positions. This step overcomes the receptive field limitations of CNN-based networks and enables the modeling of long-distance dependencies. Finally, inspired by self-attention mechanism, the Inject module is designed (shown in Fig. 6). Through an injection mechanism, it distributes the captured scene information to feature layers at different scales in the FPN, achieving adaptive fusion of scene information and mitigating the information loss issues typically encountered in traditional fusion processes. SIGM and Inject together form SGD by gathering and distributing scene-level features.

1) Scene Information Gather Module: It receives features P_3, P_4, and P_5 from various levels of the backbone network. Through the FAM, spatial alignment is performed on different levels of feature maps. Specifically, for P_5, upsampling is achieved through transposed convolution. Since P_3 contains high-resolution spatial features, in order to reduce information loss caused by pooling operations, average pooling and max pooling are applied separately [47]. The results are concatenated along the channel dimension to preserve the original information and output channels to the maximum extent, ultimately obtaining F_{align} with P_4 as the spatial reference. Subsequently, the aligned feature map F_{align} is processed through the internal Transformer block within SIEM to capture scene features S. Each block consists of a multihead attention block, a feedforward network block, and a residual connection. Drawing inspiration from LeViT [48], the channel dimensions of queries Q and keys K are set to half of values V. To further accelerate inference, layer normalization is replaced with batch normalization for convolution, and depthwise separable convolution layers are introduced in the feedforward network to minimize the impact of the transformer module on model speed. Finally, through segmentation operations, S is divided into feature maps S_3, S_4, and S_5, each with three different channel numbers.

2) Inject Module: It employs scene features S_i as queries, injecting them into the features M_i fused through FPN using a pseudo-self-attention mechanism. This process not only yields rich high-level semantic information but also facilitates selective fusion, enabling the final output feature map N_i to dynamically adjust based on the specific input, suppressing irrelevant background information while enhancing relevant contextual details.

The formula is as follows:

$$F_{\text{query}} = \text{Resize}(\text{Conv}_{1\times1}(S_i)) \tag{1}$$

$$F_{\text{key}} = \text{Sigmoid}(\text{Conv}_{1\times1}(M_i)) \tag{2}$$

$$F_{\text{value}} = \text{Conv}_{1\times1}(M_i) \tag{3}$$

$$N_i = \text{Conv}_{1\times1}(F_{\text{query}} \cdot F_{\text{key}} + F_{\text{value}}). \tag{4}$$

In the Resize operation in (1), when the size of F_{key} is larger than F_{query}, bilinear interpolation is employed for upsampling. Conversely, average pooling is utilized.

C. Omni-Dimensional C2f

In comparison to ordinary natural images, remote sensing images exhibit higher background complexity and are more susceptible to background noise interference. Additionally, remote sensing images are often captured from very high altitude, resulting in larger coverage areas. Consequently, the proportion of targets in the image tends to be small, with the majority of pixels representing background. This presents significant challenges in extracting target features from the images, especially in the shallow layers of the model. For conventional convolutions, the learning patterns across different channels are often similar, making it easy for the features extracted to be overwhelmed by background features, which dominate the majority of pixels. Some researchers have employed attention mechanisms to reassign feature weights. Wang et al. [49] utilized serial channel and spatial attention to process contextual features extracted by branches with different convolution kernels in the shallow layers of the model, enabling the model to focus more on the detailed features of the target. Gu et al. [50] proposed bidimensional attention modules (BAMs), abandoning the serial attention approach. By integrating channel dependencies and spatial relationships, they addressed attention loss issues. Additionally, their designed split and merge (SM) block captures global contextual correlations using direction-specific pooling kernels, enhancing spatial consistency while optimizing computational efficiency. However, these approaches still constrain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

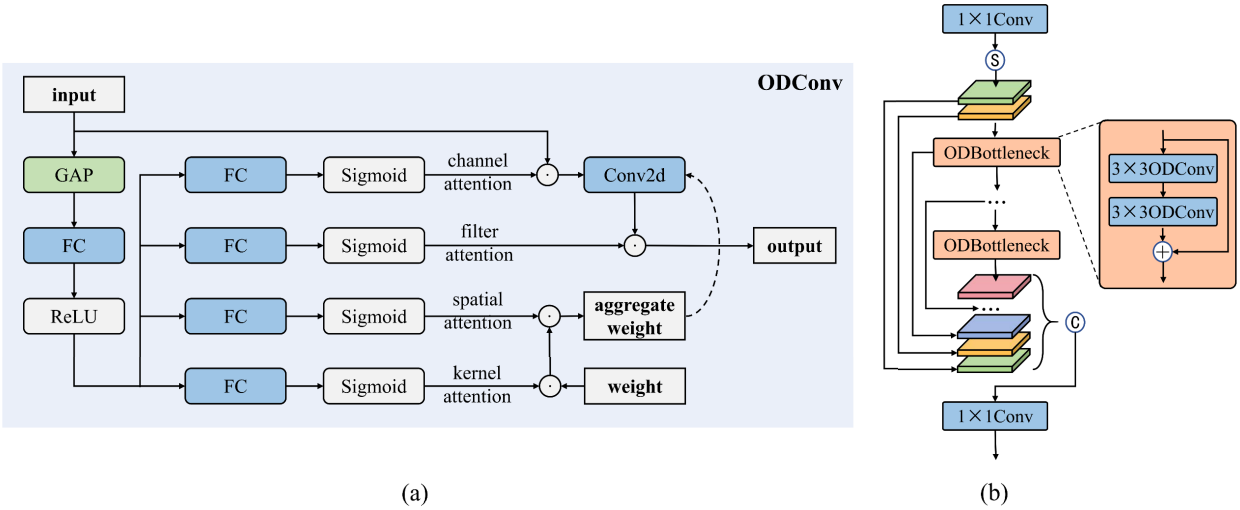


Fig. 7. Structure diagram of (a) ODConv and (b) ODC2f.

attention to the channel and spatial dimensions, and both serial and parallel strategies are unable to fully avoid attention interference problems.

To address this issue, we introduced the ODC2f structure in the shallow layers of the backbone network as a replacement for the traditional C2f module. The newly designed ODC2f structure enhances precise target localization and feature capture across all dimensions. By leveraging the complementary attention of ODConv across four dimensions and integrating the attention calculations into a single convolution operation, the structure significantly increases the granularity of attention and mitigates interference between different attention mechanisms. This improves the network's ability to accurately localize target regions and effectively capture features.

The ODConv comprises four parallel branches, each dedicated to computing attention across spatial dimensions, input channel dimensions, output channel dimensions, and kernel spatial dimensions. This enables dynamic allocation of weights to different convolutional kernels, enhancing the flexibility of convolution operations. The formula for Full-ODConv is as follows:

$$y = \left(\begin{matrix} \alpha_{\omega 1} \odot \alpha_{f1} \odot \alpha_{c1} \odot \alpha_{s1} \odot W_1 + \cdots \\ + \alpha_{wn} \odot \alpha_{fn} \odot \alpha_{cn} \odot \alpha_{sn} \odot W_n \end{matrix} \right) * x. \quad (5)$$

In (5), y represents the feature map after ODConv, and x denotes the input feature map for ODConv. W_i represents the ith convolutional kernel, and $\alpha_{\omega i}$ signifies the overall attention weight for convolutional kernels. α_{si} denotes attention weight for spatial position dimensions of convolutional kernels, and α_{ci} and α_{fi}, respectively, represent attention weights for input channel and output channel dimensions. The $\odot$ symbolizes the element-wise multiplication along different dimensions of the kernel spatial space.

The computation process of the ODConv is illustrated in Fig. 7(a). The feature map input x undergoes global average pooling, fully connected layers, and ReLU activation to be compressed into a vector of length equal to the number of channels in x. Subsequently, this vector passes through four separate branches, each containing a fully connected layer and an activation function, resulting in four attention weights: $\alpha_{\omega i}$, α_{si}, α_{ci}, and α_{fi}, corresponding to different dimensions. Finally, the convolutional filter parameters W_i are subjected to attention-weighted calculations, yielding the ultimate output y.

This article introduces a novel ODC2f module based on ODConv [see Fig. 7(b)]. The module employs an ODBottleneck, replacing the original Bottleneck in C2f. This substitution enhances the model's sensitivity to inputs, achieving autonomous learning for background suppression and target area enhancement. The ODBottleneck consists of two stacked ODConv blocks, with each block comprising a ODConv, batch normalization layer, and SiLU activation. The first ODConv block expands channels to comprehensively learn features, while the second ODConv block compresses the channels to be equal to the input channels and residually connects the result to the input.

D. SLGA Based on Prototype Learning

In the realm of target detection tasks, conventional models typically leverage sample tags at the instance level, yet they fall short of capturing relationships between diverse instances across the entire image. In reality, various object types exhibit distinctive spatial distribution patterns; for example, football fields are commonly situated within sports complexes and near ports where ships are docked, while factories tend to have more trucks, and residential areas predominantly feature cars. Consequently, we can conceptualize these relationships as scenes, implying that the probability of different objects appearing varies depending on the specific scene.

In the practical implementation of mapping instance-level labels to scene-level labels, the widespread occurrence of Incomplete manual labels poses a challenge. Additionally, within the same scene, multiple categories of ground objects may coexist, leading to a significant bias issue in scene labels generated based on one-hot encoding of the category of ground objects. As depicted in Fig. 8, within the same airport scene, the scene information provided by one-hot encoding methods varies based on the different types of parked aircraft. To address this challenge, we draw inspiration from multilevel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

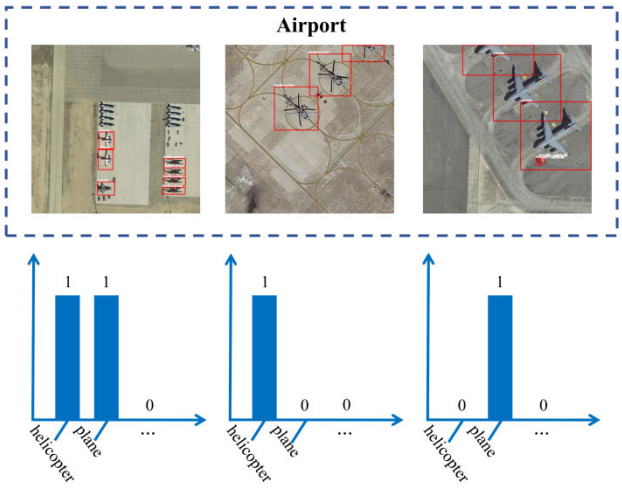


Fig. 8. Schematic of one-hot coded scene label in airport scene.

classification, leveraging human prior knowledge to categorize different ground object categories, more likely to appear in specific scenes, under the same parent category. This approach allows us to utilize other subcategories under the same parent category to infer the scene when there are not enough instances of one category. Continuing with the previous example, if we group regular airplanes and helicopters under the same parent category, labeled as "aircraft," the one-hot encoded scene vectors in Fig. 8 would remain consistent, accurately representing the scene information. This multilevel classification method not only significantly enhances the robustness of SLGAs but also introduces a novel avenue for incorporating human prior knowledge into remote sensing image object detection algorithms.

Furthermore, in remote sensing images, scenes are distributed in the form of land parcels of varying sizes. Consequently, when the bounding box encompasses two or more land parcels with different scenes, the scene vector based on one-hot encoding fails to accurately represent the distribution of distinct land parcels. This limitation hinders the model from learning the predominant scene information in the image. To effectively generate scene-level labels from instance-level labels, we have devised SLGA based on prototype learning. This method eschews one-hot encoding and, instead, derives the primary scenes by calculating the instance area proportions for each category based on the instance labels of training samples. The combination of scenes is then represented in the form of a probability distribution, yielding more adaptive scene labels.

The specific calculation process is as follows: normalize the computed class instance proportions to obtain a subclass vector, denoted as r. Subsequently, pass it through a category transformation matrix, M, to derive a parent class vector, v. Next, apply a clustering algorithm to obtain k abstracted scene vector prototypes, denoted as p_t. Finally, utilize cosine similarity as the distance metric between v and different scene prototypes, and normalize the calculated results to yield the scene label p^*

Algorithm 1 SLGA Based on Prototype Learning

Input: Training set label set L, category transformation matrix M, number of ground object categories n

1. **Initialization:** Collection of category vectors $V = \emptyset$, number of prototypes n.
2. **for** each l, v **in** L, V:
3. $r \leftarrow \emptyset$
4. **for** c **in** $[0, n)$:
5. Calculate the sum of areas for instances belonging to category c in the label set l, denoted as s.
6. $r_c \leftarrow s$
7. $v \leftarrow (r/\text{sum}(r)) \cdot M$
8. Apply the k-means++ algorithm to set V, obtaining the collection of scene prototypes Pt.
9. Calculate the cosine similarity between V and Pt to obtain the scene label P^*.
10. $P^* \leftarrow \text{softmax}(P^*)$

Output: P^*

$$M_{i,j} = \begin{cases} 1, & c_j \in C_i \\ 0, & c_j \notin C_i. \end{cases} \tag{6}$$

Equation (6) outlines the calculation method for the category transformation matrix. $M \in \mathbb{R}^{nC \times nc}$, nC denotes the number of parent categories, and nc represents the number of subcategories. $M_{i,j} = 1$ indicates that subcategory c_j belongs to parent category C_i, whereas otherwise signifies that subcategory c_j does not belong to parent category C_i. The pseudocode for the algorithm is Algorithm 1.

E. Scene-Assisted Detector Head

Traditional detection heads use multiple layers of convolution to perform regression and classification of target position and category. Although this approach enables effective detection, its limited receptive field prevents the capture of spatial long-distance dependencies, leading to certain shortcomings in fully leveraging global information. To more effectively integrate scene information to assist the model in target detection, we propose a SADHead based on the detection head of YOLOv8.

As shown in Fig. 9, we introduce a scene information encoder (SFE) into the detection head. This encoder takes the scene-level feature map S output from the SIGM. First, S undergoes processing through a 3×3 convolution layer, simultaneously enlarging the receptive field and compressing channels. Subsequently, the feature map undergoes downsampling to compress its spatial dimensions to 2×2, representing the scene information of four regions in the image. Next, the feature map S enters the channel attention module (CA) to further capture more distinctive scene features using channel context information. Finally, the feature map is flattened and input into a three-layer perceptron for regression of the scene vector $\hat{p}$.

Due to the fact that the scene vector represents the probability distribution of different specific scenes in the image,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

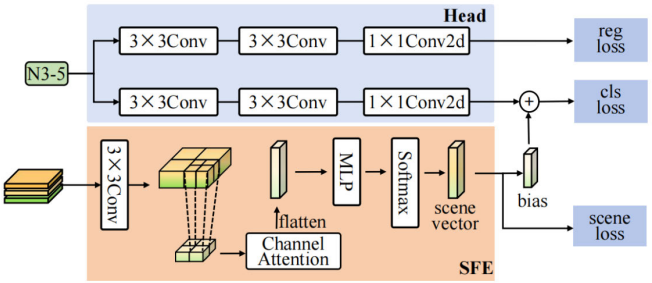


Fig. 9. SADHead structure diagram.

TABLE II
CATEGORY MAPPING TABLE FOR DOTA DATASET

Parent Category	Subcategory
C1	c5、c6
C2	c2、c4、c8、c9、c11
C3	c3、c12
C4	c7、c13、c16
C5	c1、c15
C6	c10
C7	c14

during the training phase, we employ the Kullback–Leibler (KL) divergence as the loss function for the scene vector $\hat{p}$ and the supervised label p^*

$$L_{\text{sce}} = \text{KL}(\hat{p}, p^*) = p^* \cdot (\log p^* - \log \hat{p}). \tag{7}$$

To utilize this label for assisting the SCENE-YOLO algorithm's classification, we generate a classification-enhanced bias term based on the predicted scene vector. This bias term is then added to the classification prediction output of the detection head to obtain the refined prediction output cls_{sce}. The classification-enhanced bias term corresponds to the confidence correction for each class in a specific scene. It enhances the confidence of categories belonging to that scene while suppressing the confidence of categories not belonging to that scene. This approach leverages global context information to aid in classification, reducing false positives (FPs) in detection, and enhancing overall model performance. The specific calculation formula is as follows:

$$\text{cls}, \text{box}, \hat{p} = \text{SADHead}(S, N_i) \tag{8}$$

$$\text{bias} = \hat{p} \cdot \text{Pt} \cdot M \tag{9}$$

$$\text{cls}_{\text{sce}} = \text{cls} + \text{bias}. \tag{10}$$

In (8)–(10), $\hat{p}$ represents the scene vector predicted by the model, Pt is the scene prototype matrix, and M is the category transformation matrix. The terms cls and box correspond to the predictions of the detection head for category regression.

IV. EXPERIMENTS

A. Experimental Environment

The computer processor used in this experiment is Intel[1] Xeon[1] CPU E5-2603 v4, video memory is 64 GB, GPU is NVIDIA GeForce RTX 4090, memory is 24 GB, and operating system is Ubuntu 22.04.1. Based on Python 3.10 programming experiments, the deep learning framework is Python 2.0.1.

B. Dataset

This article utilizes the DOTA dataset [51] for large-scale remote sensing object detection and the DIOR dataset annotated by the Northwestern Polytechnical University team [52] as experimental datasets.

[1]Registered trademark.

The DOTA dataset, in its version 1.5, comprises 2806 high-resolution optical remote sensing images, with dimensions ranging from 800 × 800–4000 × 4000 pixels. This dataset encompasses 16 land object categories and a total of 188 282 object instances. The 16 categories include plane (c1), baseball-diamond (c2), bridge (c3), ground track field (c4), small vehicle (c5), large vehicle (c6), ship (c7), tennis court (c8), basketball court (c9), storage tank (c10), soccer ball field (c11), roundabout (c12), harbor (c13), swimming pool (c14), helicopter (c15), and container crane (c16). To maintain uniformity in image sample size throughout this article, during training, we segment the images into subimages of 640 × 640 pixels with an overlap range of 200 pixels. Subimages without target labels are filtered out, resulting in a final set of 36 442 images.

The DIOR dataset consists of 23 463 optical remote sensing images, each sized at 800 × 800 pixels, with spatial resolutions ranging from 0.5 to 30 m. The training and validation sets include 5862 and 5863 images, respectively, accumulating a total of 190 288 object instances. The remaining images constitute the test set. This dataset encompasses 20 common land object categories: airplane (c1), airport (c2), baseball field (c3), basketball court (c4), bridge (c5), chimney (c6), dam (c7), expressway service area (c8), expressway toll station (c9), golf course (c10), ground track field (c11), harbor (c12), overpass (c13), ship (c14), stadium (c15), storage tank (c16), tennis court (c17), train station (c18), vehicle (c19), and windmill (c20).

C. Experimental Parameters of SLGA

When generating scene labels, the incorporation of prior human knowledge is necessary. This involves categorizing different object categories, more likely to appear in specific scenes, into a common parent category. This approach serves to reduce the algorithm's sensitivity to individual categories, thereby enhancing the robustness of label generation. Consequently, we manually devised mappings from parent-to-child categories for both the DOTA and DIOR datasets. These mappings were then transformed into the form of category conversion matrices. This facilitates the seamless conversion of vectors from the subcategory space to the parent category space.

For the DOTA dataset, as shown in Table II, we further divide the 16 ground object categories into 7 parent categories,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CATEGORY MAPPING TABLE FOR DIOR DATASET

Parent Category	Subcategory
C1	c1、c2
C2	c3、c4、c10、c11、c15、c17
C3	c5、c8、c9、c13
C4	c6、c16
C5	c19
C6	c18
C7	c7、c12、c14
C8	c20

with each parent category encompassing specific subcategories. These parent categories are designated as follows: vehicle (C1), sports facility (C2), highway facility (C3), port facility (C4), airport facility (C5), industrial facility (C6), and residential facility (C7).

Similarly, for the DIOR dataset, we have organized the original 20 subcategories into 8 parent categories, as outlined in Table III. These 8 parent categories are designated as follows: airport (C1), sports facility (C2), highway facility (C3), industrial facility (C4), vehicle (C5), railway facility (C6), waterside facility (C7), and windmill (C8).

D. Evaluation Metric

To assess the performance of the proposed remote sensing image object detection model, this article employs mean average precision (mAP) as evaluation metrics for the experiments.

mAP is a comprehensive index in target detection evaluation, which aims to quantify the performance of models in different categories. The calculation is based on the area under the precision-recall curve, which takes into account the overall performance of the model at different thresholds. The calculation is as

$$\text{mAP} = \int_0^1 P(R)dR. \tag{11}$$

P indicates the precision rate; R suggests the recall rate. The calculation methods for both are as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

where true positive (TP) denotes the number of targets correctly predicted as positive by the model, FP represents the count of instances where the model incorrectly predicts negative instances as positive, and false negative (FN) indicates the instances where the model mistakenly predicts positive instances as negative.

E. Experimental Parameters

In the experiments of this article, the initial learning rate is set to 0.001, weight decay is configured as 0.0005, momentum is set to 0.9, batch size is set to 12, epochs are set to 200, and the optimization method employs the adaptive moment estimation algorithm (Adam).

TABLE IV
VALIDATION BACKBONE NETWORK

Backbone	ODC2fs	Params	mAP50	mAP50-95
CSPDarkNet	0	11.16M	65.9	43.3
CSPDarkNet-A	3	11.68M	66.7	43.8
CSPDarkNet-B	3	16.52M	66.2	43.5
CSPDarkNet-C	6	17.04M	66.8	43.8

TABLE V
ABLATION EXPERIMENTS OF EACH MODULE OF THE MODEL

ODC2f	SGD	SADH	mAP50	mAP50-95
			61.3	39.6
√			62.2	40.3
	√		61.9	39.9
	√	√	63.5	41.2
√	√		62.8	40.7
√	√	√	64.1	41.6

F. Ablation Study

The ablation experiments in this article were conducted on the DOTA dataset. Initially, to select a suitable feature extraction network for the model and simultaneously validate the contribution of the ODC2f module, we used YOLOv8s as the baseline model. Based on the original CSPDarkNet, we introduced the ODC2f module at the locations of the first two C2f, the last two C2f, and all C2f, respectively, as replacements, resulting in three backbone networks: CSPDarkNet-A, CSPDarkNet-B, and CSPDarkNet-C. A comparison with CSPDarkNet was performed, and the experimental results in Table IV show that the mAP50 of the detection results for the three backbone networks with the ODC2f module reached 66.7%, 66.2%, and 66.8%, respectively. Compared to the original algorithm, these improvements were 0.8%, 0.3%, and 0.9%, indicating that early utilization of omni-dimensional attention in the model allows for rapid localization of foreground regions and suppression of further propagation of background features. Therefore, for a balanced trade off between parameter count and detection performance, CSPDarkNet-A was selected as the backbone network for the model.

To further validate the effectiveness of the proposed SGD, ODC2f, and SADHead, this article evaluates the results of experiments under the same experimental conditions, i.e., ablation experiments. YOLOv8n was chosen as the baseline model, and tests were conducted on the DOTA dataset, with experimental results presented in Table V. The results indicate that each proposed improvement contributes to a certain enhancement in the overall mAP of the final detection results.

Compared to the baseline network, the introduction of the ODC2f module led to a 0.9% and 0.7% improvement in mAP50 and mAP50-95, respectively. The SGD module improved mAP50 and mAP50-95 by 0.6% and 0.3%, respectively. Furthermore, incorporating the SADHead module

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.



Fig. 10. Experimental results on DOTA datasets.

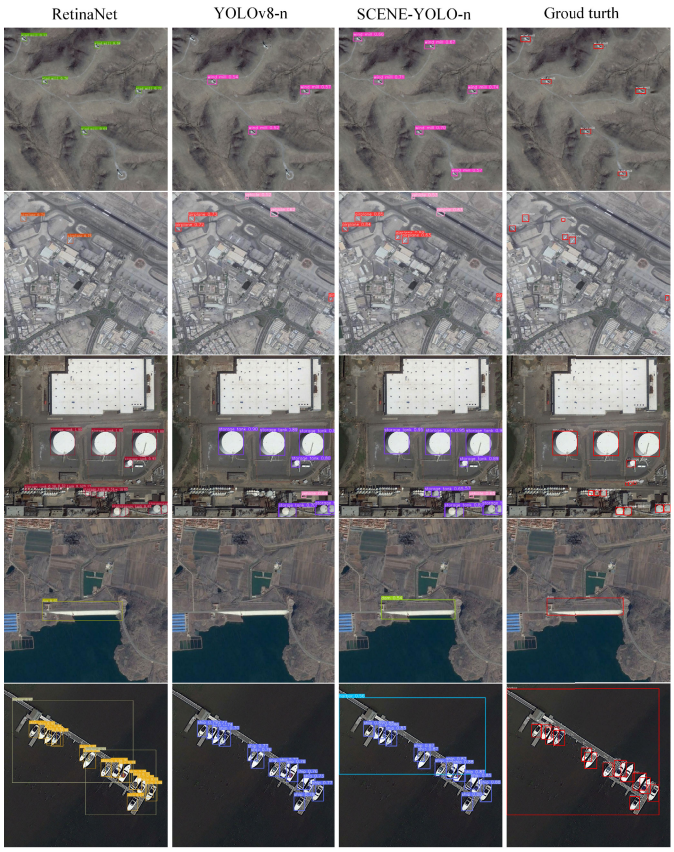


Fig. 11. Experimental results on DIOR datasets.

resulted in a notable increase, with mAP50 and mAP50-95 improving by 2.2% and 1.6%, respectively. Additionally, the combination of these modules demonstrated better performance compared to using any single module alone, affirming the effectiveness of the proposed enhancements.

To intuitively verify the performance improvement of the algorithm for remote sensing target detection, this study selected a subset of images from the DOTA and DIOR datasets. These images included targets such as vehicles, ships, harbors, football fields, bridges, airports, and windmills. The detection was performed using SCENE-YOLO-n and compared with YOLOv8-n and RetinaNet. The parameter counts for SCENE-YOLO-n and YOLOv8-n are 3.2 and 5.9 M, respectively, while RetinaNet, using ResNet50 as its backbone, has a parameter count of 32.5 M.

It can be observed that the YOLOv8-n exhibited limited detection capabilities for targets such as harbors, bridges, and airports, often resulting in missed detections. This deficiency stems from the small number of instances of these ground object categories, characterized by elongated shapes and significant angular variations. The original algorithm struggled to extract sufficient features from these targets, leading to low confidence scores. Moreover, in the presence of complex backgrounds, irrelevant background regions interfered with the feature extraction process. When targets experienced partial occlusion, it led to instances of missed or false detections, especially affecting small targets, as seen in Fig. 10 with vehicles and small storage tanks in Fig. 11. In comparison, RetinaNet has a better performance, especially on larger targets, such as airports and dams, which is related to its larger number of parameters. However, it performs poorly on dense targets compared to YOLOv8-n, with repeated detection and missed detection.

In contrast, we enhanced the model's performance by incorporating strategies such as the SGD, ODC2f, and SAD-Head. These approaches were employed to introduce richer contextual information as a reference, effectively leveraging background information for scene understanding and assisting in the detection process. This integration resulted in a significant improvement in addressing the aforementioned challenges.

As depicted in the figures, our algorithm demonstrates outstanding detection performance. With only a slight increase in the number of parameters, it achieves results comparable to RetinaNet for large targets, while also exhibiting strong detection capabilities for small and densely packed targets. The algorithm shows a more comprehensive object detection ability, effectively eliminating missed detections and balancing detection accuracy with computational cost. However, the introduction of the new scene loss does slightly affect the regression performance of the predicted bounding boxes. Nevertheless, the overall enhancement in detection capability highlights the effectiveness of the proposed strategy in improving remote sensing object detection through the use of contextual information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF OTHER METHODS ON THE DOTA DATASET

Method	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	mAP50	mAP50-95
Faster R-CNN	81.5	66.8	51.5	56.1	48.1	72.6	74.1	91.4	56.5	71.6	54.3	64.2	70.1	61.7	41.2	3.1	60.3	37.9
TOOD	83.7	66.6	50.3	62.3	51.1	76.9	75.4	91.3	56.2	75.6	51.9	60.1	68.8	62.2	43.0	1.2	61.0	38.5
RetinaNet	79.6	62.3	39.2	52.5	36.8	66.2	82.4	88.0	51.7	55.4	46.7	61.0	65.6	56.7	32.3	4.6	55.1	33.4
MSCF-CEM	88.1	70.6	49.3	48.9	66.5	82.3	87.2	92.0	61.4	74.3	56.7	61.8	66.4	71.6	44.7	5.9	64.2	43.1
CSFF	82.8	68.4	53.2	57.1	70.2	78.9	90.2	89.5	60.2	76.8	59.2	65.5	71	67.1	39.4	7.5	64.8	40.8
SRAF-Net	87.1	72.5	51.3	46.2	69.5	81.2	89.8	90.5	64.8	77.6	57.6	62.5	73.6	71.1	41.8	8.9	65.4	41.8
YOLOv5-n	82.2	61.8	41.6	46.0	64.8	79.4	90.0	92.7	54.5	74.1	48.7	59.4	71.4	61.4	32.8	6.7	60.4	39.0
YOLOv5-s	84.3	70.1	49.6	54.2	68.7	81.9	91.7	93.7	60.5	80.1	52.5	66.2	73.6	63.4	46.6	6.9	65.3	42.5
YOLOv5-l	85.3	71.5	50.9	56.4	70.7	82.7	92.9	93.4	61.6	82.6	55.6	67.4	74.3	64.9	39.3	8.6	66.1	43.2
YOLOv8-n	83.4	66.5	44.3	47.1	66.2	79.9	90.6	92.9	54.0	74.4	51.1	60.5	73.4	64.8	27.3	5.1	61.3	39.6
YOLOv8-s	85.8	70.8	48.9	54.2	69.5	83.1	92.2	94.0	60.6	81.7	54.3	64.0	75.2	67.2	45.8	7.6	65.9	43.3
YOLOv8-l	85.9	68.0	49.1	57.1	70.7	82.4	92.6	93.5	62.4	82.1	57.0	63.8	74.9	65.8	47.6	9.4	66.4	43.9
SCENE-YOLO-n	84.1	69.2	46.4	51.4	68.8	81.6	91.2	92.8	58.5	76.4	55.2	61.8	72.3	67.2	40.3	8.3	64.1	41.6
SCENE-YOLO-s	86.2	72.8	54.6	62.2	72.4	84.2	94.0	93.4	63.2	79.5	56.1	67.9	74.9	72.8	46.5	10.2	68.2	44.7
SCENE-YOLO-l	86.7	72.6	54.3	61.8	73.3	82.8	93.6	92.7	65.7	82.3	60.7	66.4	75.2	73.4	49.6	11.8	68.9	45.1

TABLE VII
COMPARISON OF OTHER METHODS ON THE DIOR DATASET

Method	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	c17	c18	c19	c20	mAP50	mAP50-95
Faster R-CNN	93.7	89.2	94.3	85.5	57.9	84.9	73.6	91.4	80.9	87.4	90.4	55.8	68.8	73.5	95.1	75.9	95.5	64.7	65.6	94.7	80.9	53.8
TOOD	93.9	91.4	94.0	85.0	55.9	85.5	72.3	90.0	79.6	84.9	88.0	59.1	67.9	72.1	93.5	73.4	94.6	62.0	61.6	93.4	79.9	55.2
RetinaNet	92.4	88.7	93.9	86.9	53.3	85.4	72.3	89.2	74.1	86.7	88.9	52.4	66.3	66.9	94.1	62.0	94.8	58.5	56.0	94.0	77.8	53.6
MSCF-CEM	97.5	91.2	95.6	90.2	64.1	91.3	65.2	94.1	87.5	87.8	91.4	71.7	70.3	92.6	93.7	92.5	95.6	63.1	72.4	94.8	85.1	62.4
CSFF	93.6	87.1	92.2	85.4	57.6	89.4	66.8	90.5	79.2	83.1	91.6	62.4	65.5	91.2	91.5	84.5	89.4	58.5	69.7	93.5	81.1	53.1
SRAF-Net	95.7	89.6	95.5	87.3	59.2	90.1	68.1	93.2	85.6	85.7	92.5	69.9	69.4	91.5	92.4	83.9	93.5	54.8	73.4	92.5	83.2	54.9
YOLOv5-n	94.9	80.2	94.0	86.5	53.0	81.4	63.9	91.1	76.7	75.1	90.2	66.2	67.6	93.2	93.3	85.7	96.0	60.6	63.2	91.5	80.2	55.5
YOLOv5-s	96.8	82.4	94.8	87.8	57.5	86.6	64.6	93.2	81.7	80.1	91.8	66.1	68.9	94.6	94.0	87.9	96.7	55.4	68.9	95.0	82.2	58.2
YOLOv5-l	97.4	77.9	95.7	88.6	57.6	86.3	56.8	91.8	82.4	82.7	90.6	65.2	69.6	95.3	93.4	89.3	97.5	60.0	71.9	96.7	82.3	59.3
YOLOv8-n	95.4	83.9	93.7	87.8	55.2	83.7	70.3	92.6	78.3	80.9	91.5	67.9	67.2	93.5	93.6	85.5	96.1	59.2	64.0	92.7	81.7	58.2
YOLOv8-s	96.9	87.5	95.4	88.7	59.2	86.6	70.1	94.3	82.4	82.9	92.9	68.1	70.4	94.7	94.6	87.8	97.3	64.4	69.2	95.7	84.0	61.0
YOLOv8-l	97.0	90.1	96.1	89.1	60.0	86.7	65.0	93.4	84.4	87.3	92.6	70.4	71.5	94.8	95.8	88.8	96.9	60.9	71.8	96.7	84.5	63.2
SCENE-YOLO-n	96.2	88.7	94.7	89.0	58.4	87.5	69.5	92.9	82.6	84.7	93.0	69.7	72.4	95.2	93.3	86.4	97.2	62.1	70.4	95.8	84.0	62.8
SCENE-YOLO-s	97.3	90.4	96.5	89.7	64.1	91.0	72.7	95.8	85.2	86.7	93.4	74.8	71.7	97.2	96.9	91.7	98.3	66.2	73.4	96.1	86.4	64.1
SCENE-YOLO-l	98.4	93.2	97.0	91.8	66.2	92.4	66.9	95.9	88.4	90.1	92.9	75.2	71.4	96.5	96.2	93.1	98.6	64.7	75.1	97.6	87.1	64.4

G. Comparison With Other Methods

To further validate the effectiveness of our proposed algorithm, we compared it with Faster R-CNN, TOOD [53], RetinaNet, YOLOv5, and YOLOv8 on the DOTA and DIOR datasets. The backbone for Faster R-CNN, TOOD, and RetinaNet was set to ResNet-50. For the YOLO series, we selected models of three different scales: *n*, *s*, and l. In addition, we also compare with some advanced remote sensing image target detection models, such as MSCF-CEM [54], CSFF [55], and SRAF-Net [41]. These models gradually increased the number of residual components and convolutional kernels, enhancing the feature extraction and fusion capabilities of the corresponding networks. However, this also led to an increase in the number of parameters and computational complexity. The training parameters for these comparative methods were referenced from their original settings and retrained in the experimental environment of this study. The final experimental results are presented in Tables VI and VII.

The comparative experimental results on the DOTA dataset demonstrate that our proposed algorithm achieves an average improvement of 3.1%, 2.3% in mAP50 and 2.5%, 1.5% in mAP50-95 compared to the different-scale series models of YOLOv5 and YOLOv8. Furthermore, our algorithm outperforms other methods, particularly excelling in the detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

of objects such as baseball field (c2), soccer field (c11), swimming pool (c14), helicopter (c15), and container crane (c16). Additionally, results on the DIOR dataset indicate the outstanding performance of the SCENE-YOLO series models in remote sensing object detection. The mAP50 for models of the n, s, and l scales reached 84.0%, 86.4%, and 87.1%, respectively, while the mAP50-95 achieved 62.8%, 64.1%, and 64.4%, respectively.

From Tables V and VI, it can be observed that RetinaNet exhibits the poorest detection performance, while the Faster R-CNN and TOOD algorithms show comparable performance, with a noticeable improvement relative to RetinaNet. SRAF-Net demonstrates exceptional performance on the DOTA dataset, which is closely related to its effective utilization of scene information. The improvements made by MSCF-CEM for multiscale objects have enhanced its detection performance for targets such as airplanes (c1) and highway service areas (c8). However, for certain categories of objects, limited sample instances, significant angle variations, and susceptibility to background interference contribute to subpar detection results. Instances of missing detections and FPs are prevalent, leading to low single-category detection accuracy. This is particularly evident for categories such as bridge (c3), ground track field (c4), soccer ball field (c11), helicopter (c15), container crane (c16) in the DOTA dataset, and bridge (c5), ship (c14), and harbor (c12) in the DIOR dataset.

Similar issues persist in the YOLO series algorithms. Our proposed algorithm effectively addresses these challenges by introducing scene information assistance. Significant improvements are particularly notable in objects strongly correlated with scene information, such as ships, bridges, cranes, and ports.

V. Conclusion

In this study, we conducted an analysis of the characteristics inherent to remote sensing images and proposed a single-stage remote sensing object detection algorithm, SCENE-YOLO, which leveraged scene information for assistance in detection. The algorithm's key innovation lies in the design of SGD, which addresses the limited receptive field of CNN networks. This module captures global contextual information to learn image scene features and facilitates the utilization of background regions that occupy a substantial portion of the image. Building upon this, a novel ODC2f module has been introduced to imbue the model with dynamic processing capabilities across four dimensions. Furthermore, the proposed SLGA and SADHead enhance the utilization of label information. By leveraging prior knowledge, these components augment the robustness of generated labels and achieve more accurate detections through the utilization of scene information.

Experimental results conducted on two publicly available remote sensing databases demonstrate that the proposed method outperforms other comparative methods in terms of mAP and significantly enhances detection performance across various land object categories. Nevertheless, the model exhibits suboptimal performance in detecting certain categories with prominent orientation variations and small targets. This limitation is attributed to the absence of considerations for features representing rotated objects and the need for a more efficient loss function. In future study, we aim to enhance the model's orientation robustness by introducing rotation-invariant features and designing specific loss functions for small and dense targets to further elevate detection performance.

References

[1] H. Cho, Y. Seo, B. V. K. V. Kumar, and R. R. Rajkumar, "A multi-sensor fusion system for moving object detection and tracking in urban driving environments," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Hong Kong, May 2014, pp. 1836–1843, doi: 10.1109/ICRA.2014.6907100.

[2] Y.-L. Lai, Y.-K. Lai, S.-Y. Shih, C.-Y. Zheng, and T.-H. Chuang, "Deep-learning object detection for resource recycling," in *Proc. J. Phys., Conf.*, Jul. 2020, vol. 1583, no. 1, Art. no. 012011, doi: 10.1088/1742-6596/1583/1/012011.

[3] M. Zurowietz and T. W. Nattkemper, "Unsupervised knowledge transfer for object detection in marine environmental monitoring and exploration," *IEEE Access*, vol. 8, pp. 143558–143568, 2020, doi: 10.1109/ACCESS.2020.3014441.

[4] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 5508605, doi: 10.1109/LGRS.2023.3309854.

[5] P. F. Felzenszwalb, R. B. Girshick, D. A. McAllester, and D. Ramanan, "Object detection with discriminatively trained part-based models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010, doi: 10.1109/TPAMI.2009.167.

[6] T. Zhang et al., "FFN: Fountain fusion net for arbitrary-oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609913, doi: 10.1109/TGRS.2023.3276995.

[7] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531916, doi: 10.1109/TGRS.2023.3335484.

[8] J. Lei, X. Luo, L. Fang, M. Wang, and Y. Gu, "Region-enhanced convolutional neural network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5693–5702, Aug. 2020, doi: 10.1109/TGRS.2020.2968802.

[9] X. Wu, W. Li, D. Hong, R. Tao, and Q. Du, "Deep learning for unmanned aerial vehicle-based object detection and tracking: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 91–124, Mar. 2022, doi: 10.1109/MGRS.2021.3115137.

[10] X. Zhang et al., "Remote sensing object detection meets deep learning: A metareview of challenges and advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 4, pp. 8–44, Dec. 2023.

[11] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019, doi: 10.1109/TGRS.2019.2930982.

[12] C. Tao, L. Mi, Y. Li, J. Qi, Y. Xiao, and J. Zhang, "Scene context-driven vehicle detection in high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7339–7351, Oct. 2019, doi: 10.1109/TGRS.2019.2912985.

[13] X. Qian, S. Gao, W. Deng, and W. Wang, "Improving oriented object detection by scene classification and task-aligned focal loss," *Mathematics*, vol. 12, no. 9, p. 1343, Apr. 2024, doi: 10.3390/math12091343.

[14] Y. Li, Y. Zhang, X. Huang, and A. L. Yuille, "Deep networks under scene-level supervision for multi-class geospatial object detection from remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 182–196, Dec. 2018, doi: 10.1016/j.isprsjprs.2018.09.014.

[15] W. Wang, L. Xing, P. Ren, Y. Jiang, G. Wang, and B. Liu, "Subspace prototype learning for few-shot remote sensing scene classification," *Signal Process.*, vol. 208, Jul. 2023, Art. no. 108976, doi: 10.1016/j.sigpro.2023.108976.

[16] F. Zhou, S. Huang, B. Liu, and D. Yang, "Multi-label image classification via category prototype compositional learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 7, pp. 4513–4525, Jul. 2022, doi: 10.1109/TCSVT.2021.3128054.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.

[17] C. Wang et al., "Gold-YOLO: Efficient object detector via gather-and-distribute mechanism," in *Proc. 37th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, New Orleans, LA, USA, 2023, pp. 51094–51112, Art. no. 2224.

[18] C. Li, A. Zhou, A. Yao, and I. China, "Omni-dimensional dynamic convolution," 2022, *arXiv:2209.07947*.

[19] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 580–587, doi: 10.1109/CVPR.2014.81.

[20] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1440–1448, doi: 10.1109/ICCV.2015.169.

[21] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017, doi: 10.1109/TPAMI.2016.2577031.

[22] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 386–397, Feb. 2020, doi: 10.1109/TPAMI.2018.2844175.

[23] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 6154–6162, doi: 10.1109/CVPR.2018.00644.

[24] C. Shi, X. Zheng, Z. Zhao, K. Zhang, Z. Su, and Q. Lu, "LSKF-YOLO: Large selective kernel feature fusion network for power tower detection in high-resolution satellite remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5620116, doi: 10.1109/TGRS.2024.3389056.

[25] G. Peng, Z. Yang, S. Wang, and Y. Zhou, "AMFLW-YOLO: A lightweight network for remote sensing image detection based on attention mechanism and multiscale feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4600916, doi: 10.1109/TGRS.2023.3327285.

[26] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 779–788, doi: 10.1109/CVPR.2016.91.

[27] C.-Y. Wang, H.-Y. M. Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, and I.-H. Yeh, "CSPNet: A new backbone that can enhance learning capability of CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Seattle, WA, USA, Jun. 2020, pp. 1571–1580, doi: 10.1109/CVPRW50498.2020.00203.

[28] Q. He, A. Xu, Z. Ye, W. Zhou, and T. Cai, "Object detection based on lightweight YOLOX for autonomous driving," *Sensors*, vol. 23, no. 17, p. 7596, Sep. 2023.

[29] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2016, pp. 21–37.

[30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007, doi: 10.1109/ICCV.2017.324.

[31] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 10778–10787, doi: 10.1109/CVPR42600.2020.01079.

[32] R. Mottaghi et al., "The role of context for object detection and semantic segmentation in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 891–898, doi: 10.1109/CVPR.2014.119.

[33] Q. Chen, Z. Song, J. Dong, Z. Huang, Y. Hua, and S. Yan, "Contextualizing object detection and classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 1, pp. 13–27, Jan. 2015, doi: 10.1109/TPAMI.2014.2343217.

[34] J. Zhang, C. Xie, X. Xu, Z. Shi, and B. Pan, "A contextual bidirectional enhancement method for remote sensing image object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4518–4531, 2020, doi: 10.1109/JSTARS.2020.3015049.

[35] Z. Zhao, J. Du, C. Li, X. Fang, Y. Xiao, and J. Tang, "Dense tiny object detection: A scene context guided approach and a unified benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5606913, doi: 10.1109/TGRS.2024.3357706.

[36] Y. Gong et al., "Context-aware convolutional neural network for object detection in VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 34–44, Jan. 2020, doi: 10.1109/TGRS.2019.2930246.

[37] T. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, Sep. 2016, pp. 1–14.

[38] S. Tian et al., "Siamese graph embedding network for object detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 602–606, Apr. 2021, doi: 10.1109/LGRS.2020.2981420.

[39] C. Chen, W. Gong, Y. Chen, and W. Li, "Object detection in remote sensing images based on a scene-contextual feature pyramid network," *Remote Sens.*, vol. 11, no. 3, p. 339, Feb. 2019, doi: 10.3390/rs11030339.

[40] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 4095–4104, doi: 10.1109/CVPR42600.2020.00415.

[41] J. Liu, S. Li, C. Zhou, X. Cao, Y. Gao, and B. Wang, "SRAF-Net: A scene-relevant anchor-free object detection network in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405914, doi: 10.1109/TGRS.2021.3124959.

[42] B. Cheng, Z. Li, B. Xu, C. Dang, and J. Deng, "Target detection in remote sensing image based on object-and-scene context constrained CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3087597.

[43] A. S. M. S. Sagar, Y. Chen, Y. K. Xie, and H. S. Kim, "MSA R-CNN: A comprehensive approach to remote sensing object detection and scene understanding," *Expert Syst. Appl.*, vol. 241, May 2024, Art. no. 122788.

[44] B. Du, Y. Huang, J. Chen, and D. Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 13435–13444, doi: 10.1109/CVPR52729.2023.01291.

[45] Z. Xie, Z. Zhang, X. Zhu, G. Huang, and S. Lin, "Spatially adaptive inference with stochastic feature sampling and interpolation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, in Lecture Notes in Computer Science. Berlin, Germany: Springer-Verlag, 2020, pp. 531–548, doi: 10.1007/978-3-030-58452-8_31.

[46] J. Xiao et al., "Tiny object detection with context enhancement and feature purification," *Expert Syst. Appl.*, vol. 211, Jan. 2023, Art. no. 118665.

[47] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2018, pp. 3–19, doi: 10.1007/978-3-030-01234-2_1.

[48] B. Graham et al., "LeViT: A vision transformer in ConvNet's clothing for faster inference," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 12239–12249, doi: 10.1109/ICCV48922.2021.01204.

[49] C. Wang, P. Dong, J. Sun, Z. Lu, K. Zhang, and W. Wan, "S-feature pyramid network and attention module for small object detection," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. Workshops (ICASSPW)*, Rhodes Island, Greece, Jun. 2023, pp. 1–5, doi: 10.1109/ICASSPW59220.2023.10193441.

[50] Y. Gu, H. Xu, Y. Quan, W. Chen, and J. Zheng, "ORSI salient object detection via bidimensional attention and full-stage semantic guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603213, doi: 10.1109/TGRS.2023.3243769.

[51] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 3974–3983, doi: 10.1109/CVPR.2018.00418.

[52] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020, doi: 10.1016/j.isprsjprs.2019.11.023.

[53] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 3490–3499, doi: 10.1109/ICCV48922.2021.00349.

[54] X. Ma, J. Dong, W. Wei, B. Zheng, J. Ma, and T. Zhou, "Remote sensing image object detection by fusing multi-scale contextual features and channel enhancement," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Gold Coast, NSW, Australia, Jun. 2023, pp. 1–7, doi: 10.1109/IJCNN54540.2023.10191739.

[55] G. Cheng, Y. Si, H. Hong, X. Yao, and L. Guo, "Cross-scale feature fusion for object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 431–435, Mar. 2021, doi: 10.1109/LGRS.2020.2975541.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.



Tao Zhao (Student Member, IEEE) received the B.S. degree in software engineering from Wuhan University of Technology, Wuhan, China, in 2022.

He is currently working toward the M.S. degree in computer technology with China University of Geosciences, Wuhan. His research interests include optical and hyperspectral remote sensing image target detection.



Ruyi Feng (Member, IEEE) received the B.S. degree in geographic information system from Hunan Normal University, Changsha, China, in 2011, and the M.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2013 and 2016, respectively.

She has been with the School of Computer Science, China University of Geosciences (Wuhan) since 2016 and is currently an Associate Professor. Her research interests include sparse representation, deep learning, hyperspectral image analysis, high-resolution remote sensing understanding, and intelligent interpretation of remote sensing imagery.



Lizhe Wang (Fellow, IEEE) received the B.E. and M.E. degree from Tsinghua University, Beijing, China, in 1998 and 2001, respectively, and the D.E. degree from the University of Karlsruhe, Germany, in 2007.

He is the Dean and "ChuTian" Chair Professor at the School of Computer Science, China University of Geosciences. His research interests include remote sensing data processing, digital earth, and big data computing.

Mr. Wang is a Fellow of the Institution of Engineering and Technology and British Computer Society and an Associate Editor of *Remote Sensing*, *International Journal of Digital Earth*, *ACM Computing Surveys*, IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, and IEEE TRANSACTIONS ON SUSTAINABLE COMPUTING. He is the recipient of Distinguished Young Scholars of NSFC, National Leading Talents of Science and Technology Innovation, 100-Talents Program of Chinese Academy of Sciences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:23 UTC from IEEE Xplore. Restrictions apply.